

Teorija polj

Vektorski operatorji

s = skalarno polje, V = vektorsko polje

grad: $\nabla s = \left(\frac{\partial s}{\partial x}, \frac{\partial s}{\partial y}, \frac{\partial s}{\partial z}\right)$

div: $\nabla \cdot \mathbf{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$

rot: $\nabla \times \mathbf{V} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix} =$

$= (Ry - Qz, Pz - Rx, Qx - Py)$

Laplacijan: $\nabla^2 s = \frac{\partial^2 s}{\partial x^2} + \frac{\partial^2 s}{\partial y^2} + \frac{\partial^2 s}{\partial z^2}$

Smerni odvod: $\frac{\partial s}{\partial l} = grad(s) \frac{\vec{l}}{|\vec{l}|}$

Prostorska geometrija

Koordinatni sistemi

Polarni: $x = r \cos \theta, \quad y = r \sin \theta \quad J = r$

Cilindrični:

$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z \quad J = r$

Sferični: $x = r \sin \phi \cos \theta, \quad y = r \sin \phi \sin \theta,$

$z = r \cos \phi \quad J = r^2 \sin \phi$

$J = \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} = \det \begin{bmatrix} x_u & x_v \\ y_u & y_v \end{bmatrix}$

Razno formule

Ločna dolžina: $s = \int_a^b |\mathbf{r}'(t)| dt =$

$= \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2} dt$

parame. točk: $\vec{r} = r_A * t(r_B - r_A)$

Odvod integrala s parametrom:

$F(x) = \int_{u(x)}^{v(x)} f(x, y) dy \Rightarrow$

$F'(x) = \int_{u(x)}^{v(x)} \frac{\partial(f(x, y))}{\partial(x)} dy +$

$+ f(x, v(x)) * v'(x) - f(x, u(x)) * u'(x)$

Naravna parameterizacija:

$|\mathbf{r}'(s)|^2 = 1, \quad s = \text{ločna dolžina}$

Tangentna premica in normalna ravnina:

$\vec{e}$ ali $\vec{n} = \dot{\vec{r}}(t_0)$

Kot med Vektorjema: $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$

Večkratni integrali

Dvojni integral:

$\iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$

Ploščina: $A = \iint_D 1 dA$

Prostornina: $V = \iiint_D f(x, y) dA$

Težišče: $\bar{x} = \frac{1}{A} \iint_D x dA, \quad \bar{y} = \frac{1}{A} \iint_D y dA$

Masa: $m = \iiint_E \rho(x, y, z) dV$

Vztrajnostni moment: $I = \iiint_E r^2 \rho dV$

Krivuljni integrali

S. polje: $\int_C f(x, y, z) ds =$

$= \int_{t_0}^{t_1} f(x(t), y(t), z(t)) * \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2} dt$

V. polje: $\int_C \mathbf{V} \cdot d\mathbf{r} = \int_C (P dx + Q dy + R dz)$

$= \int_{t_0}^{t_1} (P \dot{x} + Q \dot{y} + R \dot{z}) dt$

$d\vec{r} = \dot{\vec{r}} dt = (\dot{x} dt, \dot{y} dt, \dot{z} dt)$

Neodvisen od poti če $\vec{v}$ potencialno polje

$(rot \vec{V} = 0)$

Ploskovni integrali

S. polje: $\iint_S f dS =$

$= \iint_D f(\mathbf{r}(u, v)) \sqrt{EG - F^2} du dv$

kjer $E = \mathbf{r}_u \cdot \mathbf{r}_u, \quad F = \mathbf{r}_u \cdot \mathbf{r}_v, \quad G = \mathbf{r}_v \cdot \mathbf{r}_v$

$z = z(x, y), \quad \vec{n} = (-z_x, -z_y, 1)$

V. polje: $\iint_S \mathbf{V} \cdot d\mathbf{S} =$

$= \iint_S \mathbf{V} \cdot \vec{v} dS = \iint_D [\mathbf{V}, \mathbf{r}_u, \mathbf{r}_v] du dv$

$\vec{v} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}, \quad |\vec{r}_u \times \vec{r}_v| = \vec{n}$

Integralski izreki

Greenov, Stokesov, Gaussov :

$\oint_C (P dx + Q dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx dy$

C - rob pozitivno orientiranega sklenjenega območja D

$\oint_C \mathbf{V} \cdot d\mathbf{r} = \iint_S rot \mathbf{V} \cdot d\mathbf{S} = \iint_D (rot \vec{V}, \vec{r}_u, \vec{r}_v) du dv$

C je rob površine S, in D je projekcija na ravnino

$\iint_S \vec{V} \cdot d\mathbf{S} = \iiint_W div \vec{V} dx dy dz$

S - zaprt površina, ki je rob območja W

Kompleksna analiza

Elementarne komp. fun.

$z = x + iy; \quad |z| = \sqrt{x^2 + y^2};$

$\arg z = \arctan\left(\frac{y}{x}\right)$

$e^z = e^x (\cos y + i \sin y); \quad d\varphi = \frac{dz}{iz}$

$\log z = \log |z| + i(\arg z + 2\pi k)$

$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \cos z = \frac{e^{iz} + e^{-iz}}{2}$

$\sinh z = \frac{e^z - e^{-z}}{2}, \cosh z = \frac{e^z + e^{-z}}{2}$

Integral Kompleksne funkcije

$\int_C f(z) dz = \int_C (u + iv)(dx + idy) =$

$= \int_C (u dx - v dy) + i \int_C (v dx + u dy)$

$= \int_{t_a}^{t_b} f(z(t)) \cdot \dot{z}(t) dt$

Cauchjeva form.: $f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz$

!Integral analitične. fun. ni odvisen od poti!

Kompleksne Vrste

Taylor: $f(x) = \sum_{n=1}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$

Geometrijska, sin, cos, e vrsta:

$= \frac{1}{1 - z} = \sum_{n=0}^{\infty} z^n \quad ; |z| < 1$

$\sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} \quad ; \text{za vse } z$

$\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} \quad ; \text{za vse } z$

$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} \quad ; \text{za vse } z$

Laurent: $f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$

$a_n = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz$

Analitičnost in odvedljivost in harmoničnost

C-R pogoj: $u_x = v_y \quad u_y = -v_x \quad u - x = \frac{\partial u}{\partial x}$

$u = \int v_y dx = - \int v_x dy + C$

Pogoj harmoničnosti:

$\nabla^2 u = \nabla^2 v = 0 / w_{xx} + w_{yy} = 0$

Formule kompleksne analize

Residum: $\oint_C f(z) dz = 2\pi i \sum \text{Res}(f, z_k)$

$Res f(z) = \frac{1}{(m-1)!} * \lim_{z \rightarrow z_0} \left[\frac{d^{m-1}}{dz^{m-1}} ((z - z_0)^m f(z)) \right]$

[m]-pol n-te stopnje

L.V.: $f(z) = \sum_{n=-\infty}^{\infty} C_n (z - z_0)^n$

$C_{-1} = Res f(z) \parallel f'(z)$ za stopnje ničel

Eul. form. v $\mathbb{C}$: (realna rac. funkcija [f])

$e^{i\phi} = \cos \phi + i \sin \phi \quad z = x + iy$

$\sin(\phi) = \frac{e^{i\phi} - e^{-i\phi}}{2i}$

$\cos(\phi) = \frac{e^{i\phi} + e^{-i\phi}}{2}$

$z = e^{i\phi} |d\phi = dz / iz| \int_0^{2\pi} f(\sin(\phi), \cos(\phi)) d\phi$

$\frac{z - z^{-1}}{2i} = \sin(\phi) \mid \frac{z + z^{-1}}{2} = \cos(\phi)$

Konformne presl. (C-R), Lomljena Linearna presl. / Mobiusova presl.

Preslikava: $w = f(z) = u(x, y) + iv(x, y)$

$f(z) = \frac{az + b}{cz + d}; ad - bc \neq 0$

Integrali
Pravila integriranja

Substitucija: $\int f(g(x))g'(x)dx =$
 $= \int f(u)du, \quad u = g(x)$
Po delih: $\int u\,dv = uv - \int v\,du$
Parcialni ulomki: $\frac{P(x)}{Q(x)} = \sum \frac{A_i}{(x - a_i)^{n_i}}$

Elementarni integrali

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$
$$\int \frac{1}{x} dx = \ln|x| + C$$
$$\int e^x dx = e^x + C$$
$$\int \sin x dx = -\cos x + C$$
$$\int \cos x dx = \sin x + C$$
$$\int \frac{1}{1+x^2} dx = \arctan x + C$$
$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$$

Odvodi
Pravila odvajanja

$$(f+g)' = f' + g'$$
$$(fg)' = f'g + fg'$$
$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$
$$(f(g(x)))' = f'(g(x)) \cdot g'(x)$$

Elementarni odvodi

$$\frac{d}{dx}x^n = nx^{n-1}$$
$$\frac{d}{dx}e^x = e^x$$
$$\frac{d}{dx}\ln x = \frac{1}{x}$$
$$\frac{d}{dx}\sin x = \cos x$$
$$\frac{d}{dx}\cos x = -\sin x$$
$$\frac{d}{dx}\tan x = \sec^2 x$$
$$\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$$
$$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}}$$

Trigonometrske funkcije

$$\sin^2 x + \cos^2 x = 1$$
$$\tan x = \frac{\sin x}{\cos x}$$
$$\sin 2x = 2 \sin x \cos x$$
$$\cos 2x = \cos^2 x - \sin^2 x$$
$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$
$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$
$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$
$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

Adicijski izrek za sin,cos :

$$\sin(x + iy) = \sin x \cosh y + i \cos x \sinh y$$
$$\cos(x + iy) = \cos x \cosh y - i \sin x \sinh y$$